Leo Shaw

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Education

Carnegie Mellon University *Pittsburgh, PA*
M.S. Artificial Intelligence Engineering *Dec 2026*
Rutgers University *New Brunswick, NJ*
B.S. Mechanical Engineering; B.A. Computer Science *Dec 2024*

Technical Skills

Languages: Python, C++, Java, C, MATLAB
AI/ML: PyTorch, TensorFlow, Stable-Baselines3, OpenCV, YOLO, LLM/SLM workflows, OCR/VLM pipelines
Systems & Tools: Docker, PostgreSQL, Git, Linux, data pipelines, distributed systems, simulation tools

Projects

Graduate AI Research Lead *May 2026 - Present*
Python, Docker, PostgreSQL, data pipelines, LLMs/SLMs, OCR/VLM, RAG
- Engineered **LLM/SLM** workflows converting semiconductor PDFs, figures, tables, and telemetry into **RAG inputs** for manufacturing AI reasoning.
- Built a **Dockerized PostgreSQL** backbone for a 3.5B-5B token manufacturing AI corpus across raw documents, extracted chunks, reasoning traces, and telemetry data.
- Created **OCR/VLM** pipelines with metadata tracking and **quality gates** for 200-800 token section-aware extraction from technical documents.
- Evaluated **LLM reasoning workflows** across retrieval, chain-of-thought prompting, and tree-of-thought decomposition for complex manufacturing process tasks.

Autonomous Driving Planner with Reinforcement Learning *Jan 2026 - May 2026*
Python, PyTorch, Stable-Baselines3, CARLA, PPO, A, PID*
- Built a closed-loop **simulation stack** combining A* routing, PPO trajectory generation, PID control, and route-relative observations.
- Developed **reinforcement learning pipelines** for scenario generation, reward shaping, rollout analysis, policy benchmarking, and closed-loop policy evaluation.
- Improved task success using **PPO policy tuning**, reward shaping, and rollout analysis across highway merge and unprotected left-turn scenarios.

Reinforcement Learning for Dynamic Obstacle Navigation *Jan 2026 - May 2026*
Python, Gymnasium, Stable-Baselines3, PPO, A, NumPy, Matplotlib*
- Built a custom **simulation environment** to train PPO agents for obstacle-aware drone path planning.
- Implemented **algorithm benchmarking** by comparing A* search against learned RL policies in static and dynamic-obstacle scenarios.
- Designed **training experiments** to measure success rate, path efficiency, and adaptability across navigation conditions and obstacle layouts.

3D Perception for Occlusion Recovery *Jan 2026 - May 2026*
LaTeX, computer vision, computational imaging, neural reconstruction, optical simulation
- Studied **3D perception** and occlusion challenges affecting depth, visibility, and scene understanding.
- Analyzed **computer vision systems** combining optical simulation, neural reconstruction, and depth-aware modeling to recover occluded scene content.
- Authored a **technical research paper** covering system design, results, validation, and limitations.

Experience

Field Service Engineer I *Santa Clara, CA / Ulvac Technologies, Oct 2024 - Aug 2025*
PLC/HMI systems, valve-control software, diagnostics, system reliability, production debugging
- Delivered **production system upgrades** across PLC/HMI, RF, vacuum, and motion-handling subsystems, reducing downtime and improving process stability.
- Led **root-cause debugging** with engineering and manufacturing teams, standardizing troubleshooting procedures to accelerate fault diagnosis.
- Modified **valve-control software** and wiring logic to support upgraded tool configurations, integrate new subsystem hardware, and improve automated process control.
- Maintained tools across **high-volume** fabs through **data-driven repairs**, preventive maintenance, and calibration.